Group Project:
Modelling a forest fire using CA



Group Project

- The aim of this assignment is to explore how a CA-based simulation can be applied to represent and investigate the spread of a forest fire.
- You will be organised into groups of 4.
- You will work as a group to carry out simulation experiments to address the requirements of a fictional client who wishes you to assess the risk of forest fire to their town and potential ways of mitigating risk.

Group Project

- Your starting point for the simulation code will be the 2D Game of Life CA simulator used in the lab class in Week 5.
- This is a team project and you are expected to demonstrate your ability to work in a team.
- Detailed instructions and guidance can be found on Blackboard Ultra.

Background

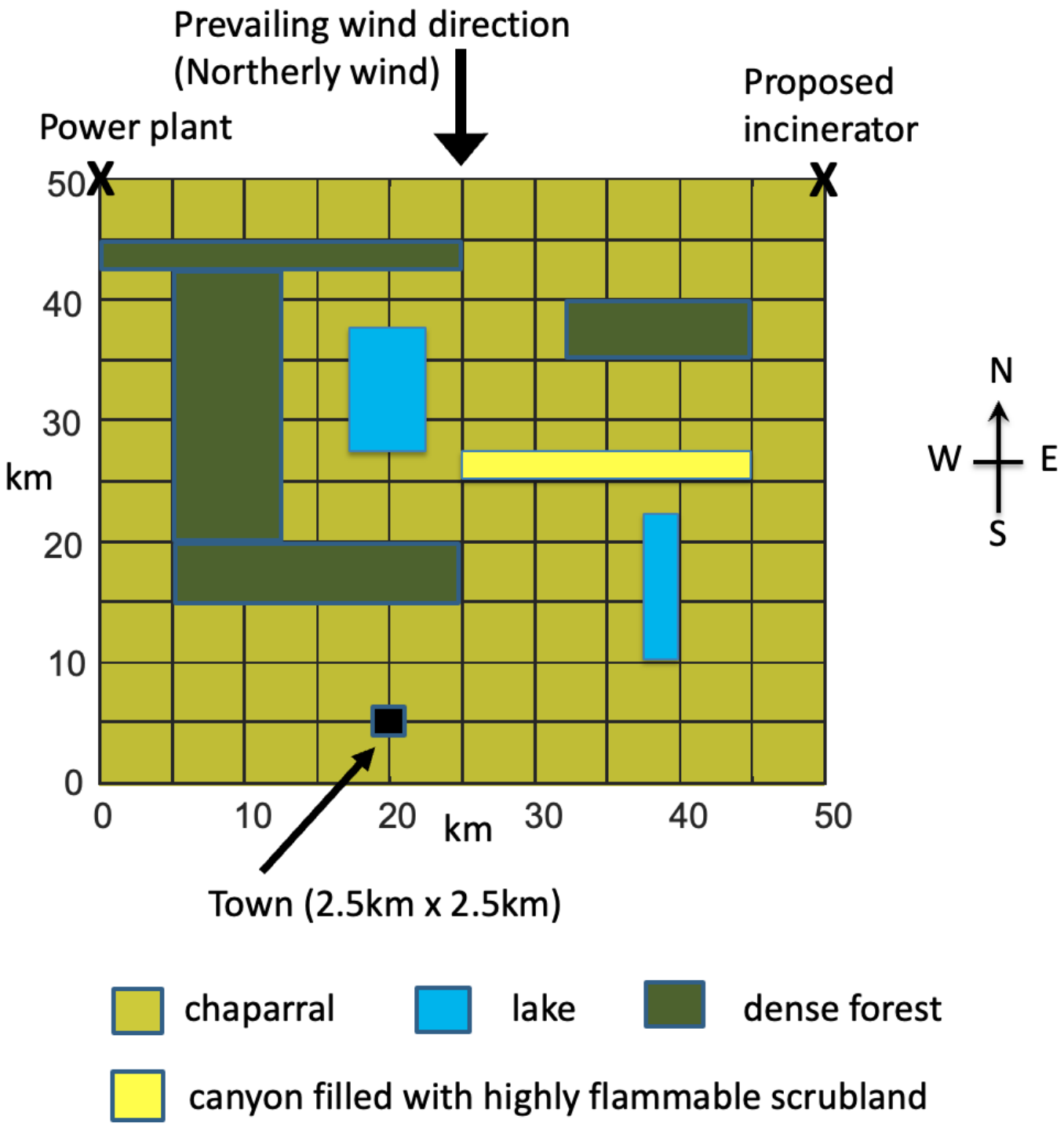
A town in the USA is located in a region prone to fire fires. Officials have become concerned about the possible **fire risk** posed to the town by:

- A **power generating plant** that is situated **to the north-west**
- A new **waste incinerator** – another potential source of ignition - has been proposed to be sited to the **north-east** of the town.

Between the town and these sites there are different types of terrain including:

- chaparral
- dense forest
- reservoir/lakes
- a canyon containing scrubland or grassland.

These have **different ignition probabilities and different burning times** and will **affect how the fire spreads.**


Prevailing wind direction
(Northerly wind)
Power plant
Proposed
incinerator
50
40
30
km
20
10
0
0
10
20
km
30
40
50
N
W
E
S
Town (2.5km x 2.5km)
chaparral
lake
dense forest
canyon filled with highly flammable scrubland

Terrain Properties

- **Dense forest** contains trees that:
  - **doesn't ignite very easily**,
  - each km^2 can **burn for up to one month**.
- **Chaparral** - fairly dense, low lying pine vegetation:
  - ignites **quite easily**,
  - each km^2 can **burn for a period of several days**.
- **Canyon** containing scrubland or grassland:
  - ignites **very easily**,
  - each km^2 **burns for a period of several hours.**
- A large reservoir/lake.

Group Project - Task

The town officials have commissioned your group of computational environmental engineers to develop a CA-based model to explore potential forest fire spread in this region. In particular, they are interested in exploring the following:

- The **relative times** for a fire starting at the power plant or starting at the incinerator to reach the town assuming a prevailing wind direction.
- How the **wind direction** affects the time it takes the fire to reach the town from the incinerator.
- The planning of **short-term intervention** in the case of a fire starting at the incinerator. Assume that there is a limited quantity of water (enough to cover only 12.5km^2) that can be dropped aerially. Use your model to explore where the best place for this drop might be.
- As a **longer term intervention**, it may be possible to extend the existing area of dense forest. How should it be extended? Are there any other longer term interventions that you could suggest/simulate?

The officials would also like you to present a brief description of the state of the art in terms of predicting the behaviour of forest fires using computational modelling.

Instructions for Wednesday's Lab (4th Nov)

- Turn up in your groups.
- Download all instructions.
  - Assignment_Overview_Slides — This powerpoint presentation
  - Assignment_Instructions — Details of what you need to do
  - Group_Project_Marking_Criteria — Details of marking criteria
  - Group_Project_Work_Plan_Template — Organise tasks/timeline between team members
  - Group_Project_Presentation_Template — Suggested presentation structure

Submission Details

- You should submit a single A4 page work plan by 5pm on Monday 9th November 2026 (Week 7).
  - **This plan will not receive a formal mark but is an important opportunity for informal feedback.**
- This assignment must be submitted by 5pm on Monday 30th November 2026 (Week 10). You must submit:
  - **A recoding of your group presentation (max 15 min).**
  - **A pdf file of your slides.**
  - **A zip file containing your group's adapted code and a readme explaining how to run it.**
- The marks will be adjusted for individual team members using BuddyCheck feedback from your peers.

Group Work: Allocation of Tasks

Think about how to allocate tasks between you and the timing of tasks:

- Searching and summarising the related literature
- Design and specification of model methodology
- Implementation of code modification/development and testing
- Specification of model assumptions and parameterisation
- Running and producing results of simulations
- Summarising work to address client's questions and critique modelling
- Finalising & recording presentation